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The impact of air pollution on the incidence of diabetes and survival among prevalent diabetes cases



Lauren A. Paul^a, Richard T. Burnett^b, Jeffrey C. Kwong^{c,d,e,f}, Perry Hystad^g, Aaron van Donkelaar^h, Li Bai^d, Mark S. Goldberg^{i,j}, Eric Lavigne^{k,l}, Ray Copes^{a,e}, Randall V. Martin^h, Alexander Kopp^d, Hong Chen^{a,b,d,e,*}

^a Department of Environmental and Occupational Health, Public Health Ontario, 480 University Ave. Suite 300, Toronto, ON M5G 1V2, Canada

^b Population Studies Division, Environmental Health Science and Research Bureau, Health Canada, Finance Bldg, 101 Tunney's Pasture Drwy, Ottawa, ON K1A 0K9, Canada

^c Public Health Ontario Laboratories, Public Health Ontario, 480 University Ave. Suite 300, Toronto, ON M5G 1V2, Canada

^d ICES, 2075 Bayview Ave. G1 06, Toronto, ON M4N 3M5, Canada

^e Dalla Lana School of Public Health, University of Toronto, 155 College St. Room 500, Toronto, ON M5T 3M7, Canada

^f Department of Family and Community Medicine, University of Toronto, 500 University Ave. 5th Floor, Toronto, ON M5G 1V7, Canada

^g College of Public Health and Human Sciences, Oregon State University, Women's Bldg, 160 SW 26th St., Corvallis, OR 97331, USA

^h Department of Physics and Atmospheric Science, Dalhousie University, Sir James Dunn Bldg, 6310 Coburg Rd., Halifax, NS B3H 4J5, Canada

ⁱ Department of Medicine, McGill University, 1001 Decarie Blvd Suite D05-2212, Montreal, QC H4A 3J1, Canada

^j Division of Clinical Epidemiology, McGill University Health Centre, 687 Pine Ave. W R4.29, Montreal, QC H3A 1A1, Canada

^k Air Health Science Division, Health Canada, 269 Laurier Ave. W A.L. 4903B, Ottawa, ON K1A 0K9, Canada

^l School of Epidemiology and Public Health, University of Ottawa, Alta Vista Campus, 600 Peter Morand Cres. Room 101, Ottawa, ON K1G 5Z3, Canada

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ABSTRACT

Purpose: Growing evidence implicates ambient air pollutants in the development of major chronic diseases and premature mortality. However, epidemiologic evidence linking air pollution to diabetes remains inconclusive. This study sought to determine the relationships between selected air pollutants (nitrogen dioxide [NO₂], fine particulate matter [PM_{2.5}], ozone [O₃], and oxidant capacity [Ox; the redox-weighted average of O₃ and NO₂]) and the incidence of diabetes, as well as the risk of cardiovascular or diabetes mortality among individuals with prevalent diabetes.

Research Design and Methods: We followed two cohorts, which included 4.8 million Ontario adults free of diabetes and 452,590 Ontario adults with prevalent diabetes, from 2001 to 2015. Area-level air pollution exposures were assigned to subjects' residential areas, and outcomes were ascertained using health administrative data with validated algorithms. We estimated hazard ratios for the association between each air pollutant and outcome using Cox proportional hazards models, and modelled the shape of the concentration-response relationships.

Results: Over the study period, 790,461 individuals were diagnosed with diabetes. Among those with prevalent diabetes, 26,653 died from diabetes and 64,773 died from cardiovascular diseases. For incident diabetes, each IQR increase in NO₂ had a hazard ratio of 1.04 (95% CI: 1.03–1.05). This relationship was relatively robust to all sensitivity analyses considered, and exhibited a near-linear shape. There were also positive associations between incident diabetes and PM_{2.5}, O₃, and Ox, but these estimates were somewhat sensitive to different models considered. Among those with prevalent diabetes, almost all pollutants were associated with increased diabetes and cardiovascular mortality risk. The strongest association was observed between diabetes mortality and exposure to NO₂ (HR = 1.08, 95% CI: 1.02–1.13).

Conclusions: Selected air pollutants, especially NO₂, were linked to an increased risk of incident diabetes, as well as risk of cardiovascular or diabetes mortality among persons with prevalent diabetes. As NO₂ is frequently used

* Corresponding author at: Population Studies Division, Environmental Health Science and Research Bureau, Health Canada, Finance Building, 101 Tunney's Pasture Drwy, Ottawa, ON K1A 0K9, Canada.

E-mail addresses: lauren.paul@oahpp.ca (L.A. Paul), rick.burnett@canada.ca (R.T. Burnett), jeff.kwong@utoronto.ca (J.C. Kwong), perry.hystad@oregonstate.edu (P. Hystad), aaron.van.donkelaar@dal.ca (A. van Donkelaar), li.bai@ices.on.ca (L. Bai), mark.goldberg@mcgill.ca (M.S. Goldberg), eric.lavigne@canada.ca (E. Lavigne), ray.copes@oahpp.ca (R. Copes), randall.martin@dal.ca (R.V. Martin), alexander.kopp@ices.on.ca (A. Kopp), hong.chen@canada.ca (H. Chen).

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as a proxy for road traffic exposures, this result may indicate that traffic-related air pollution has the strongest effect on diabetes etiology and survival after diabetes development.

1. Introduction

Diabetes mellitus is a leading public health issue worldwide, with the global prevalence of diabetes nearly doubling between 1980 and 2014 and affecting approximately 422 million adults (World Health Organization, 2016). Although many countries have developed strategies to intervene on behavioral risk factors, such as smoking and physical activity, environmental risk factors for diabetes have only been recently examined, and thus have received considerably less attention from intervention efforts (World Health Organization, 2016).

In particular, evidence from the last decade suggests that ambient air pollutants may increase rates of mortality and the incidence of major chronic diseases. Exposures to nitrogen dioxide (NO₂), fine particulate matter (aerodynamic diameter of 2.5 µg/m³ and under [PM_{2.5}]), and ozone (O₃) have been shown to contribute to morbidity and mortality in populations (Brunekreef and Holgate, 2002; Kampa and Castanas, 2008; Münzel et al., 2017b, 2017a) through pathophysiological mechanisms such as systemic inflammation and oxidative stress (Münzel et al., 2017b, 2017a). Oxidative stress is known to play a role in the etiology of diabetes, potentially leading to insulin resistance and beta cell dysfunction (Pitocco et al., 2013), as well as increasing the risk of mortality (Schöttker et al., 2015). Similarly, systemic inflammation can instigate insulin resistance, and is closely related to obesity, a major risk factor for diabetes and premature mortality (Abdelaal et al., 2017; Duncan et al., 2003; Global BMI Mortality Collaboration, 2016).

The associations between air pollutants and the incidence of diabetes have not yet been well established. Some studies have reported positive associations with NO₂ and PM_{2.5} (Bai et al., 2018; Bowe et al., 2018; Chen et al., 2013; Clark et al., 2017; Hansen et al., 2016; Krämer et al., 2010; Qiu et al., 2018), whereas others have found null associations with these pollutants (Andersen et al., 2012; Coogan et al., 2016a, 2016b, 2012; Park et al., 2015; Puett et al., 2011; Renzi et al., 2018; Weinmayr et al., 2015). Additionally, a few studies have shown an association between O₃ and the incidence of diabetes (Jerrett et al., 2017; Renzi et al., 2018).

As incident diabetes rates are increasing in young and middle-aged adults (Lipscombe and Hux, 2007), there is also increasing interest in understanding the role of air pollution in survival after the development of diabetes. A number of studies have reported that air pollution increases cause-specific mortality rates in populations, in particular cardiometabolic (cardiovascular and diabetes) mortality (Brook et al., 2013; Cakmak et al., 2019; Crouse et al., 2015; Lim et al., 2018). Some studies have also reported that those with prevalent diabetes are at increased risk of mortality with short-term exposure to air pollution (Alessandrini et al., 2016; Goldberg et al., 2013; Zanobetti et al., 2014). However, we have identified only two studies that have considered how long-term exposure to air pollution impacts survival among individuals with prevalent diabetes. These studies reported an increased risk of cardiovascular disease mortality with long-term exposure to PM_{2.5} and O₃ (Pinault et al., 2018; Pope et al., 2015).

Among persons 35–85 years of age who lived in Ontario in the years 2001–2015, we first analyzed the population who were diabetes-free at the start of 2001, and sought to estimate the associations between residential exposure to NO₂, PM_{2.5}, and O₃, as well as the combined oxidant capacity of gaseous pollutants, and the incidence of diabetes. Second, we analyzed the population with prevalent diabetes at the start of 2001, and sought to estimate the associations between the same residential air pollution exposures and cardiovascular or diabetes mortality.

2. Research design and methods

2.1. Study population

Two separate study cohorts were derived from the Ontario Population Health and Environment Cohort (ONPHEC), a retrospective closed cohort created in 2014 with current follow-up reaching 2015 (Chen et al., 2017c). ONPHEC includes all Ontario residents on April 1, 1996 who: (a) were Canadian-born, (b) were between the ages of 35 and 100, (c) had record of their birth date and sex, and (d) were registered with provincial health insurance (Appendix Definitions). The cohort was assembled by linking provincial health administrative databases housed at ICES in Toronto, Ontario. These datasets were linked using unique encoded identifiers and analyzed at ICES. The use of data in this study was authorized under section 45 of Ontario's Personal Health Information Protection Act, which does not require review by a Research Ethics Board.

To examine incident diabetes, we restricted ONPHEC to individuals who were 35–85 years of age and free of diabetes as of January 1, 2001. To examine cardiovascular and diabetes mortality among persons with prevalent diabetes, we similarly restricted ONPHEC to individuals who were 35–85 years of age, but instead restricted to those who had been identified as having diabetes prior to January 1, 2001. We further restricted both cohorts to individuals who had lived in Ontario at least 5 years prior to cohort inception. Follow-up extended to December 31, 2015.

2.2. Ascertainment of incident diabetes and mortality outcomes

Incident diabetes was ascertained using the Ontario Diabetes Database, a validated registry of diabetes in the province. The database identifies individuals who have had at least one hospital admission with a diagnosis of diabetes (ICD-9: code 250; ICD-10: codes E10–E14), or two or more physician claims related to diabetes (code 250) within a two-year period (sensitivity = 86%, specificity = 97%) (Appendix Methods for details) (Hux et al., 2002). Although the database does not distinguish between type I and type II diabetes, type II diabetes affects approximately 90% of individuals with diabetes in Canada (Doucet and Beatty, 2010). Cases of gestational diabetes are excluded. Incidence was defined as those who were diagnosed between cohort inception and the end of follow-up.

Mortality was ascertained using records from death certificates and considering the underlying cause of death. We a priori selected two causes of death for persons with prevalent diabetes as main outcomes: diabetes mortality (ICD-9: 250; ICD-10: E10–E14) and cardiovascular disease mortality (ICD-9: 401–440; ICD-10: I10–I70) (Pinault et al., 2018; Pope et al., 2015). We also considered two secondary outcomes: respiratory disease mortality (ICD-9: 460–519; ICD-10: J00–J98) and mortality from any non-accidental cause (ICD-9: < 800; ICD-10: A–R). Mortality was similarly defined as those who died between cohort inception and the end of follow-up.

2.3. Assessment of ambient concentrations of air pollution

To estimate residential exposure to PM_{2.5}, we used surface concentrations that were available each year for the period 1998–2012 at about a 1 km × 1 km resolution across Ontario. Briefly, estimates of concentration were obtained from satellite retrievals of aerosol optical depth in combination with outputs from a global chemical transport model (van Donkelaar et al., 2015). The estimates were further refined

using geographically-weighted regression with predictor variables related to land cover, elevation, and aerosol composition. Estimates correlated well with ground-based measurements at fixed-site monitors across North America ($R^2 = 0.82$, $n = 1440$) (van Donkelaar et al., 2015).

Residential exposure to NO_2 was estimated from a land-use regression model developed from Environment Canada's National Air Pollution Surveillance Network data. NO_2 concentrations were obtained from fixed-site monitoring stations across Canada, and a multiple regression model was built to estimate the 2006 exposure surface (Hystad et al., 2011). This model incorporated satellite estimates and geographic data, such as road length and population density, as predictor variables to capture background and regional variation in NO_2 (Hystad et al., 2011). The model also included predictor variables to represent fine-scale gradients in pollution concentration from important sources, such as distance to highways or major roads, to capture local-scale variation (Hystad et al., 2011). The model was validated using independent NO_2 measurements previously collected in Canadian cities. It explained 73% of the variation seen in the 2006 National Air Pollution Surveillance Network NO_2 measurements (Hystad et al., 2011).

Residential exposure to ground-level O_3 was estimated during the warm season (May–October) at a $21 \text{ km} \times 21 \text{ km}$ resolution. The estimates were derived using an optimal interpolation technique by Environment Canada, combining O_3 observations from fixed-site monitoring stations across North America with outputs from air quality prediction models (Robichaud and Ménard, 2014). The air quality models considered included the Canadian and Hemispheric Regional O_3 and NO_x System model and the Global Environmental Multi-Scale Coupled with Model of Air Quality and Chemistry model, which account for the meteorological and chemical patterns of O_3 (Robichaud and Ménard, 2014). Using this technique, Environment Canada generated annual O_3 exposure surfaces for the period 2002–2009.

Of note, NO_2 and O_3 have a close chemical relationship, whereby in the presence of sunlight, they react together and interchange molecules of oxygen (Williams et al., 2014). This reaction occurs relatively quickly, so even though levels of NO_2 and O_3 are rapidly adjusting, the sum of NO_2 and O_3 can be constant (Williams et al., 2014). Simultaneous exposure to these pollutants can cause increased oxidative stress, which may in turn pose a greater impact on health than either NO_2 or O_3 individually (Lavigne et al., 2018; Weichenthal et al., 2016; Williams et al., 2014). For these reasons, there is increasing interest in treating NO_2 and O_3 together as a metric of combined oxidant capacity (Ox) (Williams et al., 2014). Following earlier studies examining the effects of Ox (Lavigne et al., 2018; Weichenthal et al., 2016; Williams et al., 2014), we estimated this metric as a weighted average of NO_2 and O_3 using their respective redox potentials (i.e., $\text{Ox} = [1.07 * \text{NO}_2 + 2.075 * \text{O}_3]/3.145$). The greater weight placed on O_3 accounts for it being a stronger oxidizing agent than NO_2 .

We estimated pollutant exposures from 1998 (three years prior to cohort inception) until the end of follow-up in 2015. Since exposure surfaces for each pollutant were available for different periods (e.g., 1998–2012 for $\text{PM}_{2.5}$), we applied temporal adjustments to calculate annual exposure surfaces for missing years. Details on the temporal adjustment technique have been explained in detail elsewhere (Chen et al., 2017a). Annual exposure surfaces for each pollutant were linked to the cohort using annual postal code addresses (Appendix Definitions).

2.4. Covariables

Accepted or suspected risk factors for diabetes that were available from provincial health administrative databases were selected a priori to include in analyses. These included age, sex, region classification, area-level socioeconomic status (SES), and healthcare access. Similar to other large cohort studies using administrative databases (Chen et al., 2017a; Crouse et al., 2017; Lavigne et al., 2017), individual-level

covariables such as BMI, smoking, and physical activity were unavailable for the entire cohort.

Age and sex were obtained from the Registered Persons Database, which comprises all Ontario residents with provincial health insurance. All information on region classification and area-level SES for Ontario was ascertained using Canadian census records. For region classification, a north/south Ontario indicator and an urban/rural residence indicator were created (see Appendix Methods for details). For SES, we used multiple indices including quintiles of neighborhood income distribution, percentage of recent immigrants, unemployment rate, and percentage with less than high school education. These variables were obtained at the census division and dissemination area-levels (Appendix Definitions). We additionally included density of general practitioners at the census subdivision-level (Appendix Definitions) to account for the possibility that access to healthcare could influence the probability of being diagnosed with diabetes.

Furthermore, to ascertain comorbidities at the individual-level, we obtained the history of hypertension, coronary heart disease, congestive heart failure, stroke, arrhythmia, chronic obstructive pulmonary disease, asthma, and cancer at baseline using validated algorithms (Appendix Methods). These comorbidities were selected to be consistent with previous studies (Bai et al., 2018; Bowe et al., 2018; Qiu et al., 2018). We also included an ethnicity variable that was available from ICES and classifies individuals as either Chinese, South Asian, or general population (Appendix Methods).

2.5. Statistical analysis

We applied standard Cox proportional hazards models to estimate hazard ratios (HRs) and 95% confidence intervals (CIs) for the relationships between each of the four ambient air pollutants and the risk of incident diabetes, cardiovascular mortality, or diabetes mortality. Air pollution exposure was derived as 3-year moving average concentrations, allowing us to account for long-term trends in air pollution exposure. Follow-up time extended from the date of cohort entry through the date of diabetes diagnosis (incidence cohort only), death, ineligibility for provincial health insurance, or the end of follow-up, whichever occurred first. All models used time-on-study as the time scale and were stratified by an indicator for living in the Greater Toronto Area. We considered the model that adjusted for age and sex, area-level SES, region classification, and healthcare access as the fully-adjusted main model.

First, to assess the possibility of nonlinear concentration-response relationships, we applied a new class of flexible models that can capture various possible shapes of the air pollutant relationships (including linear, log-linear, and sigmoidal). Briefly, this method involves applying a function (the Shape Constrained Health Impact Function) to transform each pollutant by either a linear or log-linear function, followed by weighting the transformed value by a logistic weighting function, and considering different combinations of parameters (Nasari et al., 2016). The pollutant concentrations transformed by the function were then implemented with the fully-adjusted Cox model using standard statistical software. To account for the uncertainty in modeling, all models examined were averaged with weights corresponding to each model's log-likelihood, referred to as the ensemble model (Nasari et al., 2016). For each pollutant, we applied the method to visualize its relationship with each of the outcomes. This method has been applied to assess nonlinear relationships in previous studies of air pollution and health outcomes (Chen et al., 2013; Jerrett et al., 2017; Weichenthal et al., 2016). Further details on the methodology are available elsewhere (Nasari et al., 2016).

Second, to facilitate comparison with previous studies, we also ran Cox models with the untransformed pollutant concentrations, and adjusted sequentially by the covariables. From these models, we obtained HRs per interquartile range (IQR) increase in concentration. To assess whether there was effect modification by selected characteristics, we

performed stratified analyses by running the models in various strata defined by age group, sex, and income quintile. HRs were compared across strata using Cochran's Q statistic. To assess the robustness of the HRs, we conducted sensitivity analyses by further adjusting for ethnicity and comorbidities. To account for the possibility of residual confounding from spatial factors, we also considered a mixed-effects Cox model that included a random effect of census division (Appendix Definitions). Additionally, we ran two-pollutant and three-pollutant models of PM_{2.5}, NO₂, and O₃.

Lastly, we applied an indirect adjustment method for BMI, smoking, and physical activity using a representative auxiliary dataset that had information on these risk factors available at the individual-level (Ontario respondents to the Canadian Community Health Survey (Chen et al., 2017b, 2017a) (Appendix Methods)). The indirect adjustment uses estimated associations between the missing covariables (i.e., BMI, smoking, and physical activity) and the exposure of interest (i.e., NO₂, PM_{2.5}, O₃, or Ox) from an auxiliary dataset, along with associations between the missing covariables and the outcomes (e.g., incident diabetes) from the literature (Bhaskaran et al., 2018; Cheng et al., 2018; Cloostermans et al., 2015; Jee et al., 2010; Kyu et al., 2016; Mons et al., 2015; Pan et al., 2015; Williams, 2013), to mathematically adjust HRs for the missing covariables, while simultaneously controlling for the observed covariables (e.g., age and sex). We have used this method in several previous publications (Chen et al., 2017a, 2017b; Crouse et al., 2015). A more detailed overview of the methodology can be found elsewhere (Shin et al., 2014).

3. Results

3.1. Incident diabetes

In our cohort of ~4.8 million participants free of diabetes, 790,461 individuals were diagnosed with diabetes over the study period (Table 1). The average age of the cohort was 53 years and 52% were female. Compared with the full cohort, individuals diagnosed with diabetes were older and more likely to be male. They were also more likely to have other pre-existing comorbidities and live in areas with lower levels of income. Furthermore, the 3-year mean pollutant concentrations at baseline (in 2001) were 18.2 ppb (IQR = 13.8 ppb) for NO₂, 9.6 µg/m³ (IQR = 3.5 µg/m³) for PM_{2.5}, 46.5 ppb (IQR = 6.3 ppb) for O₃, and 37.2 ppb (IQR = 3.4 ppb) for Ox (Appendix Table 1). Pearson correlations between pollutants are shown in Appendix Table 2.

Considering concentration-response relationships first, the association between NO₂ and incident diabetes was near-linear across the entire range of concentrations, indicating proportional increases in risk per increase in concentration (Fig. 1a). Conversely, the relationship with Ox appeared to have little risk below the mid-range of concentration, above which the risk increased exponentially (Fig. 1d). For example, moving from the 1st percentile of the Ox concentration range to the 25th percentile only yielded a relative change in risk of 1.002 (95% CI: 1.001–1.003), whereas an increase from the 75th percentile to the 99th percentile yielded a relative change in risk of 1.06 (1.04–1.08) (Appendix Tables 3a and 3b). Alternatively, the concentration-response relationship for PM_{2.5} displayed some supralinear curvature, indicating that greater increases in risk were observed at lower concentrations of PM_{2.5}, while changes in risk plateaued at higher concentrations (Fig. 1b). An increase from the 1st percentile of PM_{2.5} to the 25th percentile yielded a relative change in risk of 1.02 (1.01–1.03), whereas an increase from the 75th percentile to the 99th percentile yielded a relative change in risk of 1.01 (0.99–1.03).

From the standard Cox models with untransformed pollution concentrations, each IQR increase in exposure to air pollution was associated with increased risk of incident diabetes, with the highest fully-adjusted HR of 1.04 (1.03–1.05) observed for NO₂ (Table 2). The full set of incrementally adjusted models can be found in Appendix Table 4.

The HR for NO₂ remained pronounced across all sensitivity analyses considered, although the estimate was attenuated by indirectly adjusting for BMI, smoking, and physical activity (1.03, 95% CI: 1.00–1.06) (Appendix Table 5). More modest associations were found for the three other pollutants, with fully-adjusted HRs of 1.01 (1.01–1.01) for every IQR increase in PM_{2.5}, 1.01 (1.00–1.01) for O₃, and 1.01 (1.01–1.01) for Ox (Table 2). These associations were somewhat sensitive to the alternative analyses, in particular indirect adjustment. We also found that associations with NO₂ and O₃ were robust to adjustment for other pollutants (Appendix Table 6). On the other hand, the association with PM_{2.5} was sensitive to adjustment for NO₂ and/or O₃.

Considering effect modification by selected characteristics, we found that females (female: HR = 1.07 vs. male: HR = 1.02, P = < 0.001) exhibited heightened risk of developing diabetes in association with NO₂ exposure (Fig. 2a, Appendix Table 7). For both O₃ and Ox, we found greater diabetes risk among adults aged < 65 (< 65: O₃ = 1.01, Ox = 1.02 vs. ≥ 65: O₃ = 0.99, Ox = 1.00, P = < 0.001 for both pollutants). Additionally for O₃, greater risk was observed among males. We found no evidence of effect modification for the relationship with PM_{2.5}.

Table 1

Baseline Characteristics for the Cohort without Diabetes and the Cohort with Prevalent Diabetes.

Baseline Characteristics	Cohort without Diabetes		Cohort with Prevalent Diabetes
	Full Cohort (n = 4,774,984)	Incident Diabetes Cases (n = 790,461)	(n = 452,590)
Demographic			
Age, years	53.0 ± 13.0	55.5 ± 11.9	62.8 ± 12.2
35–49	47.3	35.6	16.2
50–64	31.3	40.9	36.8
65–85	21.4	24.6	48.0
Female	52.4	47.1	46.9
Comorbidities			
Coronary heart disease	3.8	6.2	16.1
Hypertension	24.3	38.4	59.6
Congestive heart failure	1.9	2.5	10.3
Stroke	1.2	1.4	5.4
Arrhythmia	1.8	2.2	6.1
Chronic obstructive pulmonary disease	8.0	10.0	16.4
Asthma	2.5	3.2	3.4
Cancer	4.9	5.0	9.4
Neighbourhood-level socioeconomic status			
Rural residence	18.1	17.5	17.7
Northern residence	8.1	8.5	8.9
Greater Toronto Area residence	39.2	40.3	40.6
Income quintile^a			
Lowermost	16.8	19.1	22.1
Lower	19.7	21.3	22.4
Middle	20.5	20.8	20.2
Upper	20.7	19.9	18.3
Uppermost	22.2	18.8	16.7
Recent immigrants, % ^b	3.3 ± 6.1	3.6 ± 6.4	3.8 ± 6.6
Unemployed, % ^b	6.1 ± 4.7	6.3 ± 4.9	6.5 ± 5.2
> High school education, % ^b	25.8 ± 12.6	27.2 ± 12.7	28.3 ± 12.9

Values are mean ± SD or %.

^a Missingness was 0.2% for the cohort without diabetes, 0.3% for diabetes cases, and 0.3% for the cohort with prevalent diabetes.

^b From the 2001 Canadian Census at the dissemination area-level.

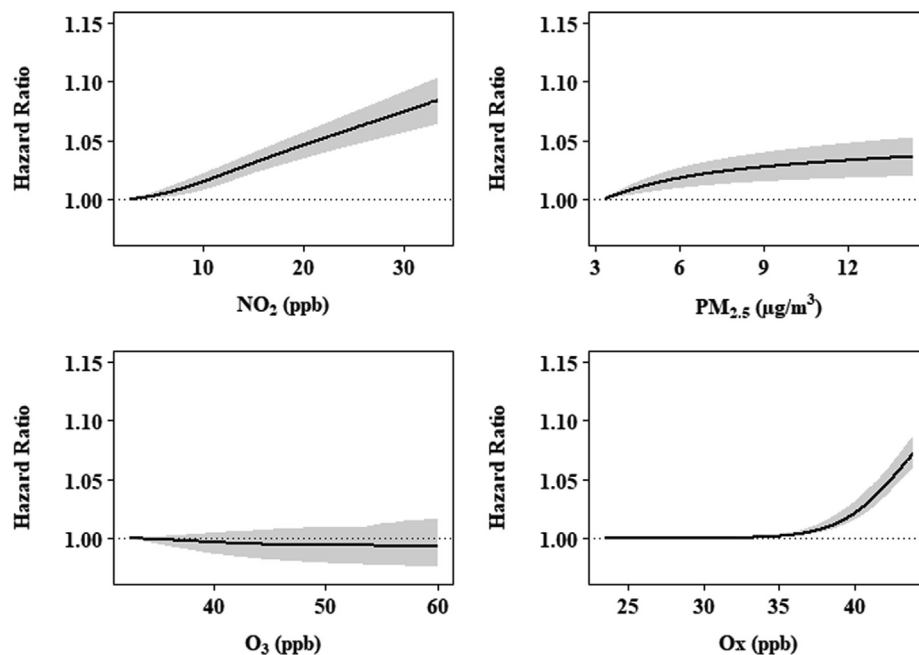


Fig. 1. Concentration-Response Functions of Ensemble Models for the Associations between NO_2 , $\text{PM}_{2.5}$, O_3 , and Ox, and the Risk of Incident Diabetes in Ontario during the Study Period (2001–2015).

Table 2

Hazard Ratios and 95% CIs for the Associations between NO_2 , $\text{PM}_{2.5}$, O_3 , and Ox, and the Risks of Incident Diabetes, Diabetes Mortality, and Cardiovascular Mortality in Ontario during the Study Period (2001–2015).

Outcome	NO_2		$\text{PM}_{2.5}$		O_3		Ox	
	HR ^a	95% CI	HR	95% CI	HR	95% CI	HR	95% CI
Incident diabetes	1.042	1.033–1.051	1.010	1.005–1.014	1.005	1.001–1.009	1.010	1.007–1.013
Diabetes mortality among persons with prevalent diabetes	1.076	1.024–1.130	1.020	0.999–1.042	1.064	1.037–1.091	1.056	1.036–1.077
Cardiovascular mortality among persons with prevalent diabetes	1.055	1.022–1.090	1.044	1.030–1.058	1.066	1.049–1.083	1.054	1.040–1.067

^a Main model including age, sex, SES, region, and healthcare access covariables. All models stratified by Greater Toronto Area residence. Hazard ratios are expressed per interquartile range (IQR) increase for each pollutant. The IQRs were as follows: for incidence, NO_2 = 13.8 ppb, $\text{PM}_{2.5}$ = 3.5 $\mu\text{g}/\text{m}^3$, O_3 = 6.3 ppb, Ox = 3.4 ppb, and for mortality, NO_2 = 14.2 ppb, $\text{PM}_{2.5}$ = 3.4 $\mu\text{g}/\text{m}^3$, O_3 = 6.4 ppb, Ox = 3.8 ppb.

3.2. Mortality among persons with prevalent diabetes

In our cohort of ~450,000 participants with prevalent diabetes, 26,653 persons died from diabetes and 64,773 died from cardiovascular diseases (Table 1, Appendix Table 8). Cohort members averaged 63 years of age and 47% were female. Similar to incident diabetes, individuals who died were older, more likely to be male, more likely to have pre-existing comorbidities, and more likely to live in areas with lower levels of income. Additionally, they were more likely to live outside the Greater Toronto Area. The 3-year mean pollutant concentrations at baseline were 16.0 ppb (IQR = 14.2 ppb) for NO_2 , 9.7 $\mu\text{g}/\text{m}^3$ (IQR = 3.4 $\mu\text{g}/\text{m}^3$) for $\text{PM}_{2.5}$, 46.9 ppb (IQR = 6.4 ppb) for O_3 , and 36.7 ppb (IQR = 3.8 ppb) for Ox (Appendix Table 1).

We observed a variety of shapes for the concentration-response relationships between air pollutants and cardiovascular and diabetes mortality. The concentration-response relationships between Ox and both mortality outcomes were near-linear across the entire exposure range (Figs. 3d and 4d). Conversely, NO_2 and diabetes mortality appeared to have little risk below the mid-range of concentration, after which risk increased linearly (Fig. 3a). An increase from the 50th percentile of NO_2 to the 75th percentile yielded a relative change in risk of 1.00 (1.00–1.01), whereas an increase from the 75th percentile to the 99th percentile yielded a relative change in risk of 1.18 (1.09–1.29) (Appendix Tables 3a and 3b). Similar to diabetes incidence, the concentration-response relationships for $\text{PM}_{2.5}$ and the mortality outcomes

displayed some supralinear curvature (Figs. 3b and 4b). This shape was also seen for the relationship between NO_2 and cardiovascular mortality (Fig. 4a). Finally, the concentration-response curves for O_3 and mortality tended towards a sigmoidal shape, with greater increases in risk observed for the early-to-midpoints of the concentration range (Figs. 3c and 4c). For example, considering cardiovascular mortality, an increase from the 25th percentile of O_3 to the 50th percentile yielded a relative change in risk of 1.05 (1.01–1.09), whereas an increase from the 50th percentile to the 75th percentile yielded a relative change in risk of 1.03 (0.97–1.11).

Standard Cox models similarly indicated that air pollutants were associated with increased risks of cardiovascular and diabetes mortality among persons with prevalent diabetes, and these associations were more pronounced than those observed for incident diabetes. For diabetes mortality, the strongest fully-adjusted associations were observed with NO_2 (1.08, 95% CI: 1.02–1.13 per IQR) and O_3 (1.06, 95% CI: 1.04–1.09) (Table 2). For cardiovascular mortality, the strongest fully-adjusted associations were with NO_2 (1.06, 95% CI: 1.02–1.09), O_3 (1.07, 95% CI: 1.05–1.08), and Ox (1.05, 95% CI: 1.04–1.07). Similarly, for non-accidental mortality we found the strongest associations to be with NO_2 (1.05, 95% CI: 1.04–1.07), O_3 (1.04, 95% CI: 1.03–1.05), and Ox (1.04, 95% CI: 1.03–1.04) (Appendix Table 9). To a lesser degree, respiratory mortality among persons with prevalent diabetes was found to be most strongly associated with O_3 , but this association was non-significant (1.02, 95% CI: 0.99–1.06) (Appendix Table 9).

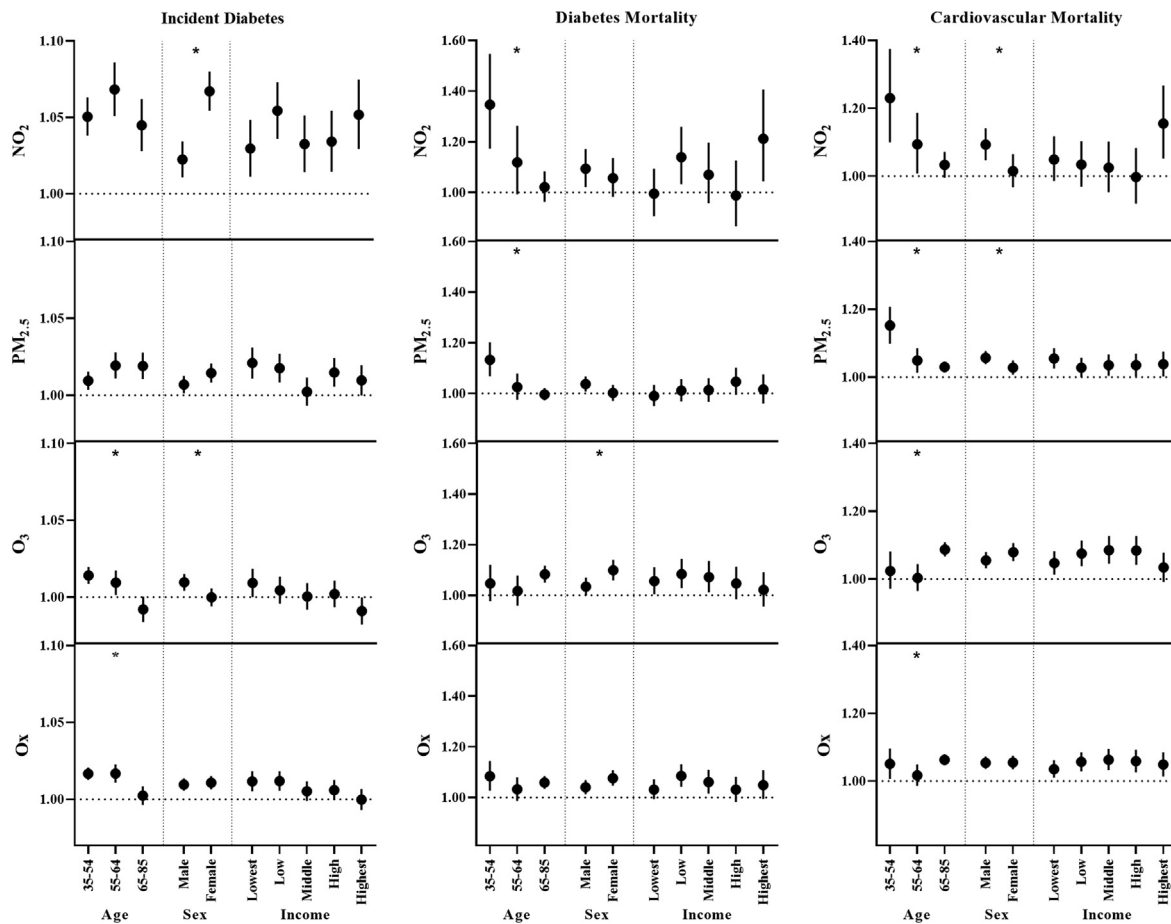


Fig. 2. Stratified Analyses by Selected Characteristics. Hazard Ratios and 95% CIs for the Associations between NO_2 , $\text{PM}_{2.5}$, O_3 , and Ox, and the Risks of Incident Diabetes, Diabetes Mortality, and Cardiovascular Mortality in Ontario during the Study Period (2001–2015). All models adjusted for age, sex, SES, region, and healthcare access, and stratified by Greater Toronto Area residence. Hazard ratios are expressed per interquartile range (IQR) increase for each pollutant. The IQRs were as follows: for incidence, $\text{NO}_2 = 13.8$ ppb, $\text{PM}_{2.5} = 3.5$ $\mu\text{g}/\text{m}^3$, $\text{O}_3 = 6.3$ ppb, Ox = 3.4 ppb, and for mortality, $\text{NO}_2 = 14.2$ ppb, $\text{PM}_{2.5} = 3.4$ $\mu\text{g}/\text{m}^3$, $\text{O}_3 = 6.4$ ppb, Ox = 3.8 ppb. Cochran's Q was used to test for heterogeneity across strata. Significant heterogeneity ($P < 0.05$) is indicated by asterisks.

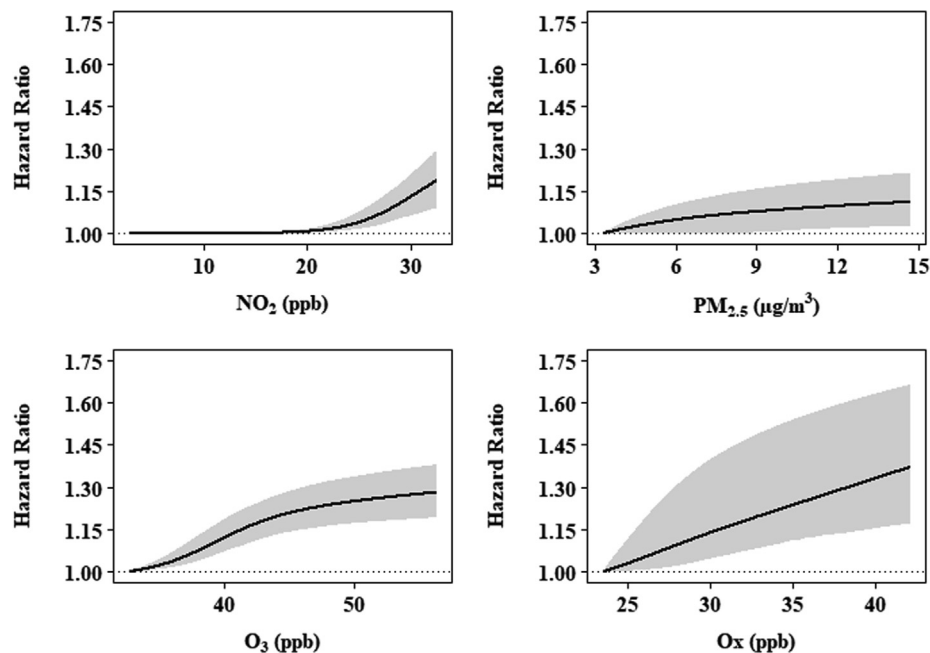


Fig. 3. Concentration-Response Functions of Ensemble Models for the Associations between NO_2 , $\text{PM}_{2.5}$, O_3 , and Ox, and the Risk of Diabetes Mortality in Ontario during the Study Period (2001–2015).

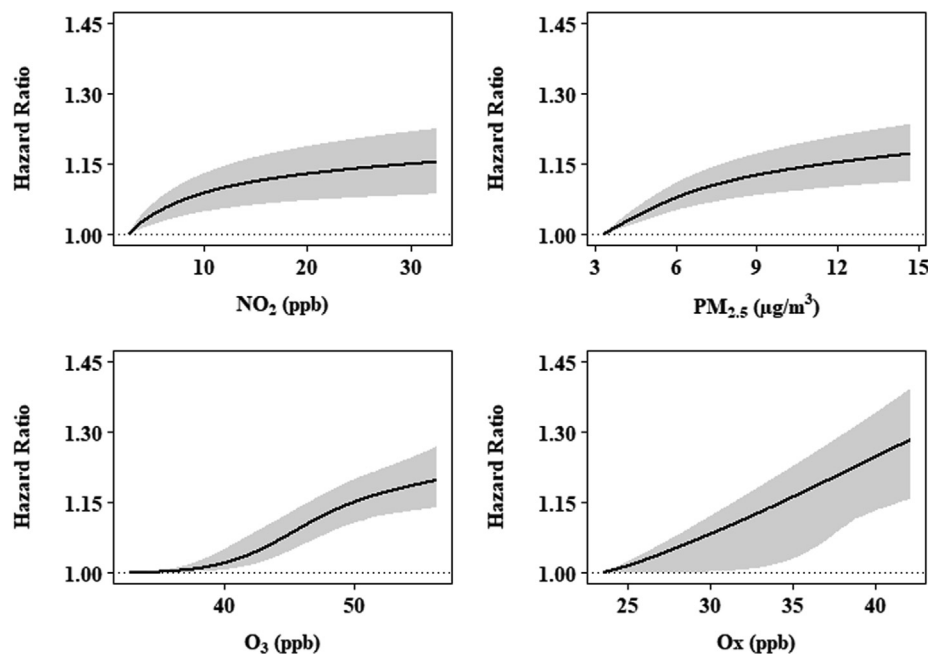


Fig. 4. Concentration-Response Functions of Ensemble Models for the Associations between NO_2 , $\text{PM}_{2.5}$, O_3 , and Ox, and the Risk of Cardiovascular Mortality in Ontario during the Study Period (2001–2015).

We also found that HRs were relatively robust in sensitivity analyses controlling for ethnicity and comorbidities, while some estimates were strengthened with the inclusion of a random effect for census division (e.g., the main model estimate for NO_2 and diabetes mortality was 1.08 (1.02–1.13), and the mixed-effect model estimate was 1.10 (1.05–1.17)) (Appendix Table 5). Similar to the incidence of diabetes, there was some attenuation after indirect adjustment with the mortality outcomes. For example, the HR for O_3 and cardiovascular mortality decreased from 1.07 (1.05–1.08) to 1.04 (1.02–1.07) after indirect adjustment. However, the NO_2 and diabetes mortality association strengthened after indirect adjustment to 1.09 (0.98–1.22). In multi-pollutant models of diabetes mortality, the associations with NO_2 and O_3 were robust to adjustment for other pollutants, and strengthened in three-pollutant models (e.g., the estimate for NO_2 increased to 1.11 (1.05–1.18)). In multi-pollutant models of cardiovascular mortality, the association with NO_2 was more sensitive to adjustment for other pollutants (e.g., the association decreased to 1.03 (1.00–1.07) in the three-pollutant model) (Appendix Table 6). Associations with $\text{PM}_{2.5}$ were sensitive to adjustment for NO_2 and/or O_3 for both mortality outcomes, as was also seen in models of incident diabetes.

For both diabetes mortality and cardiovascular mortality, adults aged < 55 with diabetes had significantly greater risk with exposure to NO_2 and $\text{PM}_{2.5}$ (e.g., for NO_2 and diabetes mortality, < 55: HR = 1.35 vs. ≥ 55 : HR = 1.04, $P = 0.001$ and for $\text{PM}_{2.5}$ and diabetes mortality, < 55: HR = 1.13 vs. ≥ 55 : HR = 1.00, $P < 0.001$) (Figs. 2b and 2c, Appendix Table 7). This result contrasted that observed for O_3 with cardiovascular mortality, where greater risk was seen for adults aged ≥ 65 with diabetes (≥ 65 : HR = 1.09 vs. < 65: HR = 1.01, interaction $P < 0.001$). Furthermore, females with diabetes had heightened risk of mortality from diabetes with exposure to O_3 (female: HR = 1.10 vs. male: HR = 1.03, interaction $P = 0.02$), while males with diabetes had greater cardiovascular mortality risk with exposure to NO_2 and $\text{PM}_{2.5}$.

4. Discussion

In our cohorts of Ontario adults, there were consistent associations linking increased incidence of diabetes to long-term residential exposure to NO_2 , and increased cardiovascular and diabetes mortality to NO_2 , O_3 , and Ox among persons with prevalent diabetes. To a lesser

degree, we found positive associations between incident diabetes and residential exposures to $\text{PM}_{2.5}$, O_3 , and Ox, but these associations were somewhat sensitive to the alternative models examined. $\text{PM}_{2.5}$ was consistently associated with cardiovascular mortality in persons with prevalent diabetes, however its association with diabetes mortality was less clear. There was also some evidence of effect modification by selected characteristics for all outcomes considered.

As NO_2 concentrations have a decay gradient with distance from roadways, and are frequently used as a proxy for road traffic exposures (HEI Panel on the Health Effects of Traffic-Related Air Pollution, 2010), our results may indicate that air pollution from traffic-related sources has the strongest effect on both diabetes development and mortality after diabetes. Our near-linear concentration-response curve for the incidence of diabetes indicates that any decrease in exposure to NO_2 could offer a significant reduction in diabetes risk, and so intervention efforts could be effective even in countries with relatively low levels of NO_2 . Interestingly, we also observed a possible threshold effect for NO_2 and the risk of diabetes mortality. These curves suggest that exposure to all levels of NO_2 can lead to diabetes development, but higher exposures are more relevant for progression to mortality from diabetes. The significant association between NO_2 and the incidence of diabetes from our study was consistent with several European studies. One study in the Ruhr district of West Germany found a HR of 1.42 (95% CI: 1.16–1.73 per 15 $\mu\text{g}/\text{m}^3$) (Krämer et al., 2010), while two studies in Denmark found HRs of 1.05 (95% CI: 0.98–1.12 per 7.5 $\mu\text{g}/\text{m}^3$) (Hansen et al., 2016) and 1.04 (95% CI: 1.01–1.07 per 4.9 $\mu\text{g}/\text{m}^3$) (Andersen et al., 2012). The latter study also found evidence of a linear dose-response relationship with NO_2 , similar to ours (Andersen et al., 2012). We further observed that females had a greater risk of diabetes with NO_2 exposure, similar to previous studies (Andersen et al., 2012; Bai et al., 2018).

Compared with NO_2 , we found $\text{PM}_{2.5}$ and O_3 to be more modestly associated with incident diabetes and mortality from diabetes. Some studies previously reported positive associations of $\text{PM}_{2.5}$ and the incidence of diabetes (Chen et al., 2013; Clark et al., 2017; Hansen et al., 2016; Krämer et al., 2010; Qiu et al., 2018), but not others (Coogan et al., 2016b, 2012; Park et al., 2015; Puett et al., 2011; Renzi et al., 2018; Weinmayr et al., 2015). Previous studies have also reported mixed results on which pollutants have the strongest effects on

cardiovascular mortality and diabetes mortality (Cakmak et al., 2019; Crouse et al., 2015; Hvidtfeldt et al., 2019; Jerrett et al., 2013; Sanyal et al., 2018). We recognize that the chemical composition and toxicity of PM_{2.5} can vary greatly over geographical regions, which may explain the inconsistent evidence to date. It is of interest for future studies to consider specific components of PM_{2.5}, and individually or simultaneously examine each component to determine which most strongly affect the risk of diabetes and/or survival after the development of diabetes.

We identified only two studies on O₃ and the incidence of diabetes, and both reported positive associations (Jerrett et al., 2017; Renzi et al., 2018). In our stratified analyses for incident diabetes, we did observe some subgroups that had more prominent associations with O₃ exposure, such as younger adults and males. One possible explanation may be that these groups engage in more outdoor activity (considering both recreational and job-related activity), thus receiving higher doses of O₃ on average. Further examining cardiovascular mortality, we found O₃ had the strongest association out of the four pollutants. This surprising result was similar to a cohort study in California that considered multi-pollutant models of PM_{2.5}, NO₂, and O₃ (Jerrett et al., 2013). However, the authors suggested that their observation may be due to the larger spatial scale of the O₃ exposure surface and resulting measurement error (Jerrett et al., 2013). Our O₃ data was similarly at a larger spatial scale than NO₂ and PM_{2.5}.

To further compare the effects of exposure to PM_{2.5}, NO₂, and O₃, we ran two-pollutant and three-pollutant models for all outcomes. Estimates of NO₂ and O₃ with incident diabetes and diabetes mortality were relatively robust to adjustment for other pollutants, whereas for cardiovascular mortality, the association with NO₂ was somewhat sensitive to adjustment for other pollutants. Associations with PM_{2.5} were sensitive to adjustment for NO₂ and O₃ for all outcomes, similar to what has been seen in previous studies of multi-pollutant models (Crouse et al., 2015; Jerrett et al., 2013). However, caution is warranted in interpreting these multi-pollutant model results as exposure data was measured at varying spatial resolutions, and thus inherently have different levels of measurement error. One previous study has suggested that the spatial resolution of O₃ captures more regional variation in air pollution, while that of NO₂ captures more local variation, and PM_{2.5} captures some of both (Crouse et al., 2015). This may be why including these three pollutants in the same model explains away associations with PM_{2.5}. Given the importance of elucidating the effects of individual air pollutants independent of other co-existing pollutants, further research is required to develop more refined exposure data and multi-pollutant modelling approaches.

To our knowledge, there are only two long-term studies that have examined air pollution and mortality among persons with prevalent diabetes. One U.S. study found that those with pre-existing diabetes had an approximate 10% increase in risk of cardiovascular mortality per 10 µg/m³ of PM_{2.5} (Pope et al., 2015). Another Canada-wide study found that those with diabetes had a 51% increase in risk for cardiovascular mortality per 10 µg/m³ increase in PM_{2.5} exposure, and a 30% increase in risk with O₃ exposure (Pinault et al., 2018). Our HRs were more conservative, and we suspect that differences are due to the study period and the way persons with prevalent diabetes were classified. The U.S. study followed cohort members from 1982 to 2004 when ambient PM_{2.5} levels were higher overall (their mean exposure was 12.6 µg/m³) and possibly had a different composition compared to PM_{2.5} in our study, although we lacked data to test this. The Canadian study identified persons with prevalent diabetes as those with diabetes indicated anywhere in their death record, while our identification was based on physician diagnoses between 1994 (the earliest year available) and 2001, prior to cohort inception. Our method may be more comprehensive in capturing all individuals who have ever received a diabetes diagnosis, but it may also lead to some misclassification if some individuals who received a prior diagnosis made subsequent lifestyle adjustments to improve their health. With type II diabetes, it is possible

to reverse the diabetes metabolic state with changes to diet, physical activity, or surgery (Taylor, 2013). If this holds true for some members of our prevalence cohort, then we may have captured more “healthy” individuals compared to the previous Canadian study, which might have diluted our effect estimates. Evidence of this may have been seen when that study considered a second Canadian cohort where persons with prevalent diabetes were identified based on self-reported diabetes at baseline (also 2001), as well as based on death certificates. With baseline identification they found an attenuated HR of 1.11 per 10 µg/m³ increase in PM_{2.5} exposure, whereas with death certificate identification they found a higher HR of 1.34 per 10 µg/m³.

Finally, previous studies have shown that Ox has stronger associations with mortality (Lavigne et al., 2018; Williams et al., 2014) and myocardial infarction (Weichenthal et al., 2016) than either NO₂ or O₃ individually. By examining our concentration-response curves, we found some evidence of a similar result for our outcomes. For example, although we did not find O₃ significantly increased the risk of incident diabetes compared to NO₂, its combined ability to cause oxidative stress with NO₂ appears to increase the risk of diabetes at higher levels of Ox. Additionally, the concentration-response curves for the associations between mortality from diabetes and NO₂ and O₃ had varying levels of risk across their concentration ranges (e.g., for NO₂ there appeared to be a risk threshold), whereas the concentration-response curve for Ox showed significant increases in risk even at the lowest concentrations. Areas with high levels of traffic-related air pollution tend to have fluctuating levels of O₃ due to both the formation of O₃ from nitrogen oxides and volatile organic compounds, and reactions between O₃ and nitrogen oxides forming NO₂. Therefore, Ox may be an important marker of the synergistic adverse effects caused by the combined exposure to NO₂ and O₃, particularly in close proximity to road traffic, leading to oxidative stress and other negative health outcomes (Williams et al., 2014). This exposure metric may warrant inclusion in future health studies of air pollution effects.

Our study has some limitations. First, as our cohort was created using administrative datasets, we did not have data on some individual-level risk factors for diabetes such as more detailed information on medical histories beyond comorbidities (e.g., treatment histories). However, we were able to indirectly adjust for three individual-level covariables (BMI, smoking, and physical activity) that were considered most likely to confound our results. Second, due to the lack of information on time-activity patterns for each individual, we were unable to obtain personal exposure for the cohort. We thus estimated air pollution exposure at the residential-level. This may have led to some non-differential exposure misclassification and possibly an underestimation of air pollution effects. In addition, we did not account for possible exposure misclassification in our health analysis, which may have an impact on our assessment of the concentration-response relationships. As more accurate exposure data become available in the future, our ability to characterize the shape of concentration-response relationships is expected to further improve. Finally, the Ontario Diabetes Database does not distinguish between type I and type II diabetes diagnoses. Given that our cohort was restricted to individuals aged 35 and older, and that type II diabetes makes up 90% of diabetes cases in Canada (Doucet and Beatty, 2010), we expect any bias to be minimal.

Our study also has a number of strengths. We employed a large, population-based cohort made up of almost the entire adult population in Ontario meeting our eligibility criteria. To our knowledge, this is the largest cohort study of air pollution and incident diabetes to date. Our study also made use of population-based databases in Ontario to ascertain the covariables and outcomes, although we recognize that diabetes is not frequently reported as the primary cause of death on death certificates, so this may have led to some underestimation of diabetes mortality (Park and Peters, 2014). Additionally, our use of time-varying measures of air pollution at relatively small spatial resolutions allowed us to more accurately capture pollution exposure for our cohort compared to previous population-based studies. Furthermore, we were able

to apply novel methodology to assess the possibility of nonlinear concentration-response forms, allowing us to gain important insight into how risk changes with varying levels of exposure to each pollutant.

5. Conclusions

Our results indicate that long-term exposure to ambient air pollution is a modifiable risk factor for morbidity and mortality related to diabetes mellitus, in particular exposure to NO₂. These associations were observed even at low concentrations of air pollution in Ontario, suggesting that continued efforts are required to improve current ambient air quality. Our findings have significant public health implications, as they support the extension of diabetes intervention efforts to include environmental risk factors that may contribute to diabetes etiology and survival after the development of diabetes.

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CRediT Authorship Contribution Statement

Lauren A. Paul: Methodology, Software, Validation, Formal analysis, Writing - original draft, Visualization, Project administration. **Richard T. Burnett:** Conceptualization, Methodology, Writing - review & editing. **Jeffrey C. Kwong:** Writing - review & editing. **Perry Hystad:** Resources, Data curation, Writing - review & editing. **Aaron Donkelaar:** Resources, Data curation, Writing - review & editing. **Li Bai:** Formal analysis, Data curation, Writing - review & editing. **Mark S. Goldberg:** Writing - review & editing. **Eric Lavigne:** Writing - review & editing. **Ray Copes:** Writing - review & editing. **Randall V. Martin:** Resources, Data curation, Writing - review & editing. **Alexander Kopp:** Data curation, Writing - review & editing. **Hong Chen:** Conceptualization, Methodology, Software, Validation, Formal analysis, Resources, Data curation, Writing - review & editing, Visualization, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interests

The authors have no conflicts of interest to declare.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envint.2019.105333>.

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